# Yuntian DENG

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## Employment

**University of Waterloo** — Waterloo, Canada
Assistant Professor, Cheriton School of Computer Science — Aug 2024–Present

**Vector Institute** — Toronto, Canada
Faculty Affiliate — Aug 2024–Present

**Harvard University** — Remote
Associate in Computer Science — Jun 2024–Present

**NVIDIA** — Toronto, Canada
Visiting Professor (manager: Yejin Choi) — Nov 2024–Nov 2025

**Allen Institute for Artificial Intelligence** — Seattle, WA
Postdoc (advised by Yejin Choi) — Jul 2023–Jul 2024

## Research Interests

ProgramAsWeights (PAW) and neural programs; neural world models and interactive systems; implicit chain-of-thought reasoning and internalized computation; real-world LLM use and open research infrastructure.

## Education

**Harvard University** — Cambridge, MA
Ph.D. in Computer Science — May 2023
Advisors: Alexander Rush and Stuart Shieber

**Carnegie Mellon University** — Pittsburgh, PA
Master of Language Technologies — Aug 2016
Advisor: Eric Xing

**Tsinghua University** — Beijing, China
Bachelor of Engineering, Department of Automation — Jul 2014

## Awards

Laude Institute Slingshots (NeuralOS) — Jun 2026
Google Research Award (co-PI) — Apr 2026
Argonne National Lab Impact Award — Jan 2023
University of Chicago Rising Stars in Data Science — Nov 2022
**ACM Gordon Bell Special Prize for COVID-19 Research** — Nov 2022
**NVIDIA Graduate Fellowship** — Dec 2021
Microsoft Turing Academic Program (implicit chain-of-thought research) — Dec 2021
Harvard Certificates of Distinction in Teaching — Spring 2019, Fall 2020, Fall 2021
Twitch Fellowship Finalist — Jan 2021
DAC 2020 Best Paper Award — Jul 2020
**Baidu Fellowship** — 2019
French American Doctoral Exchange Program (French Embassy, 10 US laureates) — Jun 2018
ACL 2017 Best Demonstration Paper Runner-Up — Aug 2017

## Research Contributions

**ProgramAsWeights (PAW).** Compiles English function specifications into compact neural programs that run locally; its public hub contains 5,486 community-built programs.
*Representative work:* Program-as-Weights: A Programming Paradigm for Fuzzy Functions (*NeurIPS 2026*)
*Project:* programasweights.com

**NeuralOS.** Uses a neural generative model to simulate an interactive operating system end to end; its public deployment has 50,000+ visits.
*Representative work:* NeuralOS: Towards Simulating Operating Systems via Neural Generative Models (*ICLR 2026*)
*Project:* neural-os.com

**Implicit Chain of Thought.** Introduced implicit chain-of-thought reasoning as a research direction and helped catalyze its growth, including by co-organizing the Latent & Implicit Thinking Workshop at ICLR 2026, which accepted 78 papers.
*Representative works:* Implicit Chain of Thought Reasoning via Knowledge Distillation (*arXiv 2023*); From Explicit CoT to Implicit CoT (*arXiv 2024*)
*Demo:* A 12-layer GPT-2 achieves 99.5% accuracy when multiplying two 20-digit numbers without emitting intermediate reasoning

**WildChat.** Provides 4.8 million real user–ChatGPT conversations and an open visualizer; it is among the AI Observatory's largest and most detailed datasets and is used by OpenAI and Anthropic for safety evaluation.
*Representative work:* WildChat: 1M ChatGPT Interaction Logs in the Wild (*ICLR 2024* **Spotlight**)
*Dataset:* WildChat-4.8M-Full on Hugging Face; *Visualizer:* wildvisualizer.com

## Selected Media Coverage

### Interviews, Features, and Expert Commentary

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|---|---|
| Science News: Can AI reason without words? A small model puts the idea to the test (expert commentary) | Sep 22, 2026 |
| User Mag (Taylor Lorenz): Vibe Coders Now Have Their Own Language | Aug 26, 2026 |
| The Canadian Press: Researchers develop tool to help courts spot AI-generated evidence | Dec 5, 2025 |
| NZZ: What people really ask ChatGPT (English translation) | Nov 30, 2024 |
| TechCrunch: Why is ChatGPT so bad at math? | Oct 2, 2024 |
| The Washington Post: What do people really ask chatbots? | Aug 4, 2024 |

### Coverage of Coauthored Research

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|---|---|
| The Washington Post homepage (The 7): New data shows people are getting personal with AI | Aug 18, 2026 |
| MIT Technology Review: We still don't know how people are really using AI | Aug 18, 2026 |

### University Expert Q&As

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|---|---|
| University of Waterloo: What is "Claudish"? | Sep 8, 2026 |
| University of Waterloo: Why ChatGPT struggles with math | Oct 17, 2024 |

## Publications and Preprints

* Equal contribution; † Equal advising.

50 **Yuntian Deng**, Pengyu Nie, Stuart Shieber. Compile by Training: Turning Natural-Language Specifications into Local Neural Functions. *EMNLP 2026 Demo*

49 Wentao Zhang, Shariyar Murtaza, Junaid Bhatti, Utkarsh Soni, Yifan Nie, Eugene Wen, **Yuntian Deng**. Control-Data Flow Separation: Stable Prompt Optimization in Multi-Agent LLMs. *EMNLP 2026 Findings*

48 Wentao Zhang*, Xuanhe Pan*, Han Zhou*, Yang Lu†, **Yuntian Deng**†. Interactive Training 2: Auditable Control Plane for Live Model Training. *arXiv 2026*

47 Wentao Zhang*, Liliana Hotsko*, Woojeong Kim*, Pengyu Nie, Stuart Shieber, **Yuntian Deng**. Program-as-Weights: A Programming Paradigm for Fuzzy Functions. *NeurIPS 2026*

46 Kelly McConvey, Sajad Ebrahimi, Nima Jamali, Jalehsadat Mahdavimoghaddam, Matina Mahdizadeh Sani, Maksym Taranukhin, Wentao Zhang, Jaquelyn Burkell, **Yuntian Deng**, Karen Eltis, Maura Grossman, Vered Shwartz, Ebrahim Bagheri. A Benchmark & Dataset for Detecting AI-Manipulated Visual Evidence in the Court System. *CIKM 2026*

45 Liliana Hotsko, Yinxi Li, **Yuntian Deng**, Pengyu Nie. Code2LoRA: Hypernetwork-Generated Adapters for Code Language Models under Software Evolution. *arXiv 2026*

44 Sajad Ebrahimi, Nima Jamali, Bardia Shirsalimian, Kelly McConvey, Wentao Zhang, Jalehsadat Mahdavimoghaddam, Maksym Taranukhin, Maura R. Grossman, Vered Shwartz, **Yuntian Deng**, Ebrahim Bagheri. DetectZoo: A Unified Toolkit for AI-Generated Content Detection Across Text, Audio, and Image Modalities. *arXiv 2026*

43 Shayne Longpre, Anka Reuel, Dayeon Ki, Zhiping Zhang, Cathy Mengying Fang, Chuanyang Jin, Jennifer Mickel, Cedric Deslandes Whitney, Niloofar Mireshghallah, Victor Ojewale, Megan Richards, Ariel N. Lee, Alex Pentland, Tianshi Li, **Yuntian Deng**, Sara Hooker, Mykel Kochenderfer, Sanmi Koyejo. The AI Observatory: A Public Measure of Real-World AI Use. *NeurIPS 2026 Evaluations & Datasets Track*

42 David Acuna, Chao-Han Huck Yang, **Yuntian Deng**, Jaehun Jung, Ximing Lu, Prithviraj Ammanabrolu, Hyunwoo Kim, Yuan-Hong Liao, Yejin Choi. Long Grounded Thoughts: Synthesizing Visual Problems and Reasoning Chains at Scale. *ICML 2026*

41 Yinxi Li, **Yuntian Deng**, Pengyu Nie. TokDrift: When LLM Speaks in Subwords but Code Speaks in Grammar. *ACL 2026*

40 Luke Rivard, Sun Sun, Hongyu Guo, Wenhu Chen, **Yuntian Deng**. NeuralOS: Towards Simulating Operating Systems via Neural Generative Models. *ICLR 2026*

39 Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, Junteng Liu, **Yuntian Deng**, Yizhe Zhang, Junxian He. Learn to Reason Efficiently with Adaptive Length-Based Reward Shaping. *ICLR 2026*

38 Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, **Yuntian Deng**, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, Sara Hooker. The Leaderboard Illusion. *NeurIPS 2025 Datasets & Benchmarks Track*

37 Wentao Zhang, Yang Young Lu, **Yuntian Deng**. Interactive Training: Feedback-Driven Neural Network Optimization. *EMNLP 2025 Demo*

36 Wentao Zhang, Woojeong Kim, **Yuntian Deng**. From Chat Logs to Collective Insights: Aggregative Question Answering. *EMNLP 2025* **Oral**

35 Xiaoyan Bai, Itamar Pres, **Yuntian Deng**, Chenhao Tan, Stuart Shieber, Fernanda Viégas, Martin Wattenberg, Andrew Lee. Why Can't Transformers Learn Multiplication? Reverse-Engineering Reveals Long-Range Dependency Pitfalls. *arXiv 2025*

34 Yifan Zong, **Yuntian Deng**, Pengyu Nie. Mix-of-Language-Experts Architecture for Multilingual Programming. *LLM4Code 2025*

33 Bill Yuchen Lin, **Yuntian Deng**, Khyathi Chandu, Faeze Brahman, Abhilasha Ravichander, Valentina Pyatkin, Nouha Dziri, Ronan Le Bras, Yejin Choi. WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild. *ICLR 2025* **Spotlight**

32 Zhangchen Xu, Fengqing Jiang, Luyao Niu, **Yuntian Deng**, Radha Poovendran, Yejin Choi, Bill Yuchen Lin. Magpie: Alignment Data Synthesis from Scratch by Prompting Aligned LLMs with Nothing. *ICLR 2025*

31 Jinjie Ni, Fuzhao Xue, Xiang Yue, **Yuntian Deng**, Mahir Shah, Kabir Jain, Graham Neubig, Yang You. MixEval: Deriving Wisdom of the Crowd from LLM Benchmark Mixtures. *NeurIPS 2024*

30 **Yuntian Deng**, Wenting Zhao, Jack Hessel, Xiang Ren, Claire Cardie, Yejin Choi. WildVis: Open Source Visualizer for Million-Scale Chat Logs in the Wild. *EMNLP 2024 Demo*

29 Wenting Zhao, Tanya Goyal, Yu Ying Chiu, Liwei Jiang, Benjamin Newman, Abhilasha Ravichander, Khyathi Chandu, Ronan Le Bras, Claire Cardie, **Yuntian Deng**, Yejin Choi. WildHallucinations: Evaluating Long-form Factuality in LLMs with Real-World Entity Queries. *arXiv 2024*

28 **Yuntian Deng**, Yejin Choi, Stuart Shieber. From Explicit CoT to Implicit CoT: Learning to Internalize CoT Step by Step. *arXiv 2024*

27 Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, **Yuntian Deng**. WildChat: 1M ChatGPT Interaction Logs in the Wild. *ICLR 2024* **Spotlight**

26 John X. Morris, Chandan Singh, Alexander Rush, Jianfeng Gao, **Yuntian Deng**. Tree Prompting: Efficient Task Adaptation without Fine-Tuning. *EMNLP 2023*

25 **Yuntian Deng**, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, Stuart Shieber. Implicit Chain of Thought Reasoning via Knowledge Distillation. *arXiv 2023*

24 Maxim Zvyagin*, Alexander Brace*, Kyle Hippe*, **Yuntian Deng***, Bin Zhang, Cindy Orozco Bohorquez, Austin Clyde, Bharat Kale, Danilo Perez-Rivera, Heng Ma, Carla M. Mann, Michael Irvin, J. Gregory Pauloski, Logan Ward, Valerie Hayot-Sasson, Murali Emani, Sam Foreman, Zhen Xie, Diangen Lin, Maulik Shukla, Weili Nie, Josh Romero, Christian Dallago, Arash Vahdat, Chaowei Xiao, Thomas Gibbs, Ian Foster, James J. Davis, Michael E. Papka, Thomas Brettin, Rick Stevens, Anima Anandkumar, Venkatram Vishwanath, Arvind Ramanathan. GenSLMs: Genome-scale language models reveal SARS-CoV-2 evolutionary dynamics. *The International Journal of High Performance Computing Applications 2023* **ACM Gordon Bell Special Prize for COVID-19 Research**

23 **Yuntian Deng**, Noriyuki Kojima, Alexander Rush. Markup-to-Image Diffusion Models with Scheduled Sampling. *ICLR 2023* (**Demo**: huggingface.co/spaces/yuntian-deng/latex2im)

22 Richa Rastogi, Yair Schiff, Alon Hacohen, Zhaozhi Li, Ian Lee, **Yuntian Deng**, Mert R. Sabuncu, Volodymyr Kuleshov. Semi-Parametric Inducing Point Networks and Neural Processes. *ICLR 2023*

21 **Yuntian Deng**, Volodymyr Kuleshov, Alexander Rush. Model Criticism for Long-Form Text Generation. *EMNLP 2022*

20 Anton Bakhtin*, **Yuntian Deng***, Sam Gross, Myle Ott, Marc'Aurelio Ranzato, Arthur Szlam. Residual Energy-Based Models for Text. *JMLR 2021*

19 Justin Chiu*, **Yuntian Deng***, Alexander Rush. Low-Rank Constraints for Fast Inference in Structured Models. *NeurIPS 2021*

18 **Yuntian Deng**, Alexander Rush. Sequence-to-Lattice Models for Fast Translation. *EMNLP 2021 Findings*

17 Keyon Vafa, **Yuntian Deng**, David Blei, Alexander Rush. Rationales for Sequential Predictions. *EMNLP 2021* **Oral**

16 **Yuntian Deng**, Alexander Rush. Cascaded Text Generation with Markov Transformers. *NeurIPS 2020*

15 Thierry Tambe, En-Yu Yang, Zishen Wan, **Yuntian Deng**, Vijay Janapa Reddi, Alexander Rush, David Brooks, Gu-Yeon Wei. Algorithm-Hardware Co-Design of Adaptive Floating-Point Encodings for Resilient Deep Learning Inference. *DAC 2020* **Best Paper Award**

14 **Yuntian Deng**, Anton Bakhtin, Myle Ott, Arthur Szlam, Marc'Aurelio Ranzato. Residual Energy-Based Models for Text Generation. *ICLR 2020*

13 Zachary Ziegler*, **Yuntian Deng***, Alexander Rush. Neural Linguistic Steganography. *EMNLP 2019* **Oral** (**Demo**: steganography.live)

12 **Yuntian Deng**, David Rosenberg, Gideon Mann. Challenges in End-to-End Neural Scientific Table Recognition. *ICDAR 2019*

11 Anton Bakhtin, Sam Gross, Myle Ott, **Yuntian Deng**, Marc'Aurelio Ranzato, Arthur Szlam. Real or Fake? Learning to Discriminate Machine from Human Generated Text. *arXiv 2019*

10 **Yuntian Deng***, Yoon Kim*, Justin Chiu, Demi Guo, Alexander Rush. Latent Alignment and Variational Attention. *NeurIPS 2018*

9 Sebastian Gehrmann, **Yuntian Deng**, Alexander Rush. Bottom-Up Abstractive Summarization. *EMNLP 2018*

8 **Yuntian Deng**, Anssi Kanervisto, Jeffrey Ling, Alexander Rush. Image-to-Markup Generation with Coarse-to-Fine Attention. *ICML 2017* (**Demo**: im2markup.yuntiandeng.com)

7 Pengtao Xie, **Yuntian Deng**, Yi Zhou, Abhimanu Kumar, Yaoliang Yu, James Zou, Eric P. Xing. Learning Latent Space Models with Angular Constraints. *ICML 2017*

6 Guillaume Klein, Yoon Kim, **Yuntian Deng**, Jean Senellart, Alexander M. Rush. OpenNMT: Open-Source Toolkit for Neural Machine Translation. *ACL 2017 Demo* **Best Demonstration Paper Runner-Up**

5 Zichao Yang, Zhiting Hu, **Yuntian Deng**, Chris Dyer, Alex Smola. Neural Machine Translation with Recurrent Attention Modeling. *EACL 2017*

4 Xuezhe Ma, Yingkai Gao, Zhiting Hu, Yaoliang Yu, **Yuntian Deng**, Eduard Hovy. Dropout with Expectation-linear Regularization. *ICLR 2017*

3 Hao Zhang, Zhiting Hu, **Yuntian Deng**, Mrinmaya Sachan, Zhicheng Yan, Eric P. Xing. Learning Concept Taxonomies from Multi-modal Data. *ACL 2016*

2 Pengtao Xie, **Yuntian Deng**, Eric P. Xing. Diversifying Restricted Boltzmann Machine for Document Modeling. *KDD 2015*

1 Zhiting Hu, Poyao Huang, **Yuntian Deng**, Yingkai Gao, Eric P. Xing. Entity Hierarchy Embedding. *ACL 2015*

## Research Funding

**Laude Institute Slingshots** — USD 45,000 in-kind support — Jun 2026
*Project:* NeuralOS

**Google Research Award** — co-PI; USD 25,000 unrestricted gift and USD 75,000 in Google Cloud credits — Apr 2026
*Project:* LLM-Guided Interactive Training of Biomedical Foundation Models on TPUs

**CIFAR Canadian AI Safety Institute – AI Safety Solution Network** — co-PI; CAD 700,000 shared network funding — 2025–2027
*Project:* Safeguarding Courts from Synthetic AI Content

**Canada Foundation for Innovation John R. Evans Leaders Fund** — PI; CAD 244,846 total project value, including CAD 96,000 CFI contribution — 2024–2028
*Project:* Progressive Internalization of Skills for Advanced Language Model Reasoning

**Digital Research Alliance of Canada, Resources for Research Groups** — PI; RRG 2025 No. 5275, estimated in-kind value CAD 61,165; RRG 2026 ID 3797, estimated in-kind value CAD 28,772 — Jul 2025–Apr 2027
*Project:* Neural OS: Simulating Operating Systems using Diffusion Models

**Mitacs Business Strategy Internship** — PI; CAD 10,000 — 2025
*Project:* Decentralized Electric Microgrid Optimization by Reinforcement Learning

**Manulife Research Grant** — PI; CAD 30,000 — 2024–2026
*Project:* End-to-End Optimization of LLM-Based Multi-Agent Systems via Text Differentiation

**NSERC Discovery Grant** — PI; RGPIN-2024-05178; CAD 41,000/year + CAD 12,500 Discovery Launch Supplement — Aug 2024–Apr 2029
*Project:* Multi-Agent Communication and Collaboration for Specialized Large Language Models

**University of Waterloo Starter Grant** — CAD 150,000 — Aug 2024–Aug 2029

## Professional Service

### Editorial and Leadership

Co-Chief Technical Officer, ACL Rolling Review (ARR) Dec 2025–Present

Co-lead technical infrastructure and operations across successive review cycles; ARR received 17,087 submissions in May 2026 alone.

- Develop and operate reviewer, area-chair, and senior-area-chair matching workflows with conflict-of-interest, workload, seniority, and resubmission constraints; lead AC/SAC matching.
- Improve contributor-qualification and registration systems, and develop capacity-analysis and recruitment tooling to support reciprocal reviewing.
- Implement assignment and access-control audits, operational safeguards, and cross-cycle repairs; coordinate technical priorities with editors and the technical team.

Action Editor, Transactions of the Association for Computational Linguistics (TACL) 2025–Present

Co-Chair, AAAI-27 Undergraduate Consortium 2027

Mentor Chair, ICLR 2026

Workshop Organizer, Latent & Implicit Thinking – Going Beyond CoT Reasoning, ICLR Apr 27, 2026

### Area Chair and Senior Program Committee Roles

**Area Chair:** ICLR (2024–2026), ICML (2026), NeurIPS (2026), ACL Rolling Review (2024–2026), COLM (2025–2026), EMNLP Demo (2026), EACL Demo (2027), LREC (2026), EMNLP GEM Workshop (2023), and IJCNLP–AACL (Language Modeling, 2025).

**Senior Program Committee:** AAAI 2027.

### Selected Reviewing and Selection Service

Reviewer, *Nature* 2026

Reviewer, Schmidt Sciences Science of Trustworthy AI 2026

**Journals:** TACL, TMLR, Computational Linguistics, ACM AI Letters, ACM Computing Surveys, Foundations and Trends in Signal Processing, IEEE TNNLS, IEEE TIFS, ACM Transactions on Interactive Intelligent Systems, ACM Transactions on Information Systems, and Journal of Computer Science and Technology.

**Conferences:** ICML, NeurIPS, ICLR, ACL, ACL Rolling Review, EMNLP, AAAI, COLM, COLING, AACL, EACL, NAACL, ACL Demo, NAACL Demo, EMNLP Demo, and various workshops and reproducibility challenges.

Proposal Reviewer, Mitacs

Selection Committee Member, Vector Scholarship in Artificial Intelligence 2026–2027

### Conference Service

Oral Session Moderator, ICLR 2024, 2026

Oral Session Chair, EMNLP 2024, 2026

Co-host, ICLR Mentorship Session Apr 23, 2026

Earlier conference volunteer service: ACL (2020–2021), EMNLP (2021), and ICML (2016, 2020).

### Reviewing Recognition

Top 10% Reviewer, NeurIPS 2020

Best Reviewer (Top 10%), ICML 2021

## Teaching Experience

**Waterloo**

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|---|---|
| Instructor, CS 486/686: Introduction to Artificial Intelligence | Spring 2026 |
| Instructor, CS 486/686: Introduction to Artificial Intelligence | Spring 2025 |
| Guest Lecturer, Computational Linguistics (Freda Shi) | Feb 11, 2025 |
| Instructor, CS 886: Topics in Language Models | Fall 2024 |

**Harvard University**

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|---|---|
| Head TA, Intro to Computational Linguistics and NLP (Stuart Shieber) | Fall 2020; Fall 2021 |
| TA, Natural Language Processing (Alexander Rush) | Jan 2019–May 2019 |
| Guest Lecturer, Natural Language Processing – Translation (Alexander Rush) | Feb 13, 2019 |
| TA, Advanced Machine Learning (Alexander Rush) | Aug 2017–Dec 2017 |

**Cornell University**

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|---|---|
| Course Materials Contributor, Break Through Tech AI (Alexander Rush) | Aug 2021 |

**Carnegie Mellon University**

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| TA, Probabilistic Graphical Models (Matthew Gormley, Eric Xing) | Jan 2016–May 2016 |
| TA, Graduate Machine Learning (Ziv Bar-Joseph, Eric Xing) | Aug 2015–Dec 2015 |

## Community Service and Outreach

**Vision 2026 STEAM Conference: Beyond ChatGPT: What Else Can AI Become?** Oct 14, 2026
Outreach seminar for high-school students, co-presented with Bing Yan. Wilfrid Laurier University, Brantford.

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|---|---|
| Co-Treasurer, Board of Directors, Bright Starts Early Learning Centre | May 2025–May 2027 |
| Childcare Chair, NAACL 2022 Diversity & Inclusion Committee | Apr–Jul 2022 |
| Leader, Harvard English Language Table | Sep 2021–Jan 2023 |
| Mentor, Harvard Women in CS NLP Reading Group | Sep 2020–Sep 2023 |
| Volunteer Instructor of AP CS, Revere High School | Sep 2020–May 2021 |

## Research Internships

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|---|---|
| NVIDIA (Anima Anandkumar, Weili Nie, Arash Vahdat, Chaowei Xiao) | May–Dec 2022 |
| Facebook AI Research (Marc'Aurelio Ranzato, Arthur Szlam) | May–Dec 2019 |
| Bloomberg CTO Office (David Rosenberg, Gideon Mann) | Jan–Aug 2017 |
| UCSD (Charles Elkan) | Jul–Sep 2013 |

## Talks

**Invited Talks and Seminars**

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|---|---|
| University of Waterloo AI Reading Group: ProgramAsWeights | Oct 2026 |
| Apple MLR Internal Reading Group: NeuralOS | Aug 2026, Remote |
| Vector Endless Summer School: NeuralOS | Jun 2026, Remote |
| NSERC CREATE Responsible AI consortium: NeuralOS | Mar 2026, Remote |
| Mila/McGill NLP Reading Group: Implicit Reasoning in Language Models | Oct 2025, Remote |
| University of Cambridge Language Technology Lab: Implicit Chain-of-Thought: Internalizing Reasoning in Language Models | Oct 2025, Remote |

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|---|---|
| Google DeepMind: Implicit Reasoning in Language Models | May 2025, Remote |
| Colin Raffel Group: Implicit Reasoning in Language Models | Mar 2025, Toronto, Canada |
| University of Oklahoma: Implicit Chain of Thought Reasoning | Jan 2025, Remote |
| NYU: Implicit Chain of Thought Reasoning | Nov 2024, New York, NY |
| Google DeepMind: Implicit Chain of Thought Reasoning | Jul 2024, Remote |
| Queen's University: Implicit Chain of Thought Reasoning | Feb 2024, Remote |
| UBC: Implicit Chain of Thought Reasoning | Jan 2024, Vancouver, Canada |
| AI2 Mosaic: WildChat | Sep 2023, Seattle, WA |
| AI2 Semantic Scholar: Structure Modeling in Language Models | Aug 2023, Seattle, WA |
| NYU AI School 2023: Natural Language Processing | Jun 2023, New York, NY |
| AI2 Mosaic: Structure Modeling in Language Models | Jun 2023, Seattle, WA |
| UMass Amherst: Structural Coherence in Text Generation | May 2023, Remote |
| Georgia Tech: Structural Coherence in Text Generation | Apr 2023, Atlanta, GA |
| Harvard Kempner Institute: Structural Coherence in Text Generation | Mar 2023, Remote |
| University of Alberta: Structural Coherence in Text Generation | Mar 2023, Edmonton, Canada |
| TTIC: Structural Coherence in Text Generation | Feb 2023, Chicago, IL |
| University of Waterloo ECE: Structural Coherence in Text Generation | Feb 2023, Remote |
| Cornell Seminar in NLU: Structural Coherence in Text Generation | Feb 2023, New York, NY |
| University of Waterloo CS: Structural Coherence in Text Generation | Jan 2023, Waterloo, Canada |
| University of Chicago Rising Stars in Data Science: Model Criticism for Long-Form Text Generation | Nov 2022, Chicago, IL |
| Princeton NLP: Model Criticism for Long-Form Text Generation | Nov 2022, Princeton, NJ |
| OpenAI: Residual Energy-based Models for Text Generation | Apr 2021, Remote |
| Baidu: Cascaded Text Generation with Markov Transformers | Jun 2020, Remote |
| LinkedIn: Residual Energy-based Models for Text Generation | Mar 2020, Remote |
| Wayfair: Neural Encoder-Decoder Models | Nov 2019, Boston, MA |
| FAIR NLP Meeting: Energy-Based Models for Text Generation | Aug 2019, New York, NY |
| The French National Center for Scientific Research: Variational Attention | Jul 2018, France |
| Tencent Social Network Group TSAIC: OpenNMT | Dec 2017, Shenzhen, China |

**Conference and Workshop Presentations**

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|---|---|
| EMNLP 2021 Findings: Sequence-to-Lattice Models for Fast Translation | Nov 2021, Remote |
| NeurIPS 2020: Cascaded Text Generation with Markov Transformers | Dec 2020, Remote |
| ICLR 2020: Residual Energy-based Models for Text Generation | Apr 2020, Remote |
| EMNLP 2019: Neural Linguistic Steganography | Nov 2019, Hong Kong |
| NeurIPS 2018: Latent Alignment and Variational Attention | Dec 2018, Montreal, Canada |
| Association for Machine Translation in the Americas: OpenNMT | Mar 2018, Boston, MA |
| Open-Source Neural Machine Translation Workshop: Image-to-Text | Mar 2018, Paris, France |
| ICML: Image-to-Markup Generation | Aug 2017, Sydney, Australia |

## Panels

**KAIST at ICML** — Jul 6, 2026, Seoul
Panelist on vision, multimodal AI, and agents.

**Vector Institute's Remarkable 2025** — Mar 2025, Toronto
Panelist on emerging questions in NLP.

**ACL Mentorship: Sharing Learnings & Identifying Research Directions** — Dec 2023, Singapore
Panelist on promising research directions.

**ACL Mentorship: Preparing Your Grad School Application** — Oct 2023, Remote
Mentorship panelist on graduate-school applications.

**NAACL 2022: Balancing Professional and Family Commitments** — Jul 2022, Remote
Co-host on balancing professional and family commitments.

## Graduate Examination and Milestone Service

All service listed below is at the University of Waterloo; students' advisers are shown in parentheses unless my supervisory role is noted.

### Thesis Supervision and Examination Committees

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|---|---|
| Cong Wei — PhD Thesis Committee (Wenhu Chen and Jimmy Lin) | Oct 21, 2026 |
| Weiming Ren — PhD Thesis Committee (Wenhu Chen and Jimmy Lin) | Oct 19, 2026 |
| Dongfu Jiang — PhD Thesis Committee (Wenhu Chen) | Oct 5, 2026 |
| Henry Lin — MMath Thesis Supervision | Aug 17, 2026 |
| Han Zhou — MMath Thesis Committee (Yang Lu) | Aug 17, 2026 |
| Guilherme Campos — MASc Thesis Committee (Wojciech Golab and Tahsin Reza) | Aug 12, 2026 |
| Benjamin Schneider — MMath Thesis Committee (Wenhu Chen and Florian Kerschbaum) | Aug 11, 2026 |
| Arthur Chen — MMath Thesis Committee (Victor Zhong) | Aug 4, 2026 |
| Ruotian Wu — MMath Thesis Committee (Pascal Poupart) | Aug 4, 2026 |
| William Loh — PhD Thesis Committee (Pascal Poupart) | Jul 21, 2026 |
| Xueguang Ma — PhD Thesis Committee (Jimmy Lin) | Jul 15, 2026 |
| Hosna Oyarhoseini — MMath Thesis Committee (Jimmy Lin and Amir-Hossein Karimi) | Jul 14, 2026 |
| Xiang Li — PhD Comprehensive-II Committee (Jian Zhao and Kate Larson) | Jun 26, 2026 |
| Justin Xu — MMath Thesis Committee (Pascal Poupart) | May 2026 |
| Saarang Agarwal — MMath Thesis Committee (Mei Nagappan and Pengyu Nie) | Apr 2026 |
| Weiming Ren — PhD Comprehensive-II Committee (Wenhu Chen and Jimmy Lin) | Apr 2026 |
| Dongfu Jiang — PhD Comprehensive-II Committee (Wenhu Chen) | Apr 3, 2026 |
| Cong Wei — PhD Comprehensive-II Committee (Wenhu Chen) | Mar 2026 |
| Yuansheng Ni — MMath Thesis Committee (Wenhu Chen) | Nov 2025 |
| Xueguang Ma — PhD Comprehensive-II Committee (Jimmy Lin) | Oct 2025 |
| Allen Zhang — MMath Thesis Committee (Anita Layton) | Aug 2025 |
| Carter Blair — MMath Thesis Committee (Kate Larson and Edith Law) | Jun 2025 |
| Achint Soni — MMath Thesis Committee (Sirisha Rambhatla and Charles Clarke) | May 2025 |
| Chi-Chung Cheung — MMath Thesis Committee (Anita Layton) | Dec 2024 |
| William Loh — PhD Thesis Proposal Committee (Pascal Poupart) | Aug 2024 |

### PhD Seminar Service

Faculty review and approval of Waterloo's required PhD seminars; each seminar requires at least two faculty sign-offs.

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|---|---|
| Dongfu Jiang — 2 seminars (Wenhu Chen) | Jul 15 and Aug 5, 2026 |
| Cong Wei — 2 seminars (Wenhu Chen) | Jul 24 and 27, 2026 |
| Weiming Ren — 2 seminars (Wenhu Chen) | Jul 15 and 22, 2026 |
| William Loh — 3 seminars (Pascal Poupart) | Mar 13; Jun 5 and 15, 2026 |
| Ronak Pradeep — 1 seminar (Jimmy Lin) | Oct 9, 2025 |

## Open Source Projects

ProgramAsWeights (PAW) (230 GitHub stars; 5,486 community-built neural programs)

English ↔ Claudish (two local neural functions compiled with ProgramAsWeights; 1 million+ views and 17,000+ likes on X)

NeuralOS (202 GitHub stars; 50,000+ demo visits)

WildChat-4.8M-Full (909 total Hugging Face likes across five AllenAI versions)

OpenAIWatch (101,000 visits since 2025; 87,000 in 2026 YTD)

WildVisualizer (3.1 million visits since 2025; 2.0 million in 2026 YTD)

Internalize CoT Step by Step (208 GitHub stars)

Implicit Chain of Thought (149 GitHub stars)

WildChat Chatbots (o1, o1-mini, GPT-4, and ChatGPT; 1,616 total Hugging Face likes)

Cascaded Generation (130 GitHub stars)

Neural Linguistic Steganography (14,200 demo visits since 2025; 214 GitHub stars)

Variational Attention (327 GitHub stars)

Image-to-Markup Generation (1,300+ GitHub stars)

OpenNMT-py (7,000+ GitHub stars)

OpenNMT (archived) (2,400+ GitHub stars)

Attention OCR (1,100+ GitHub stars)